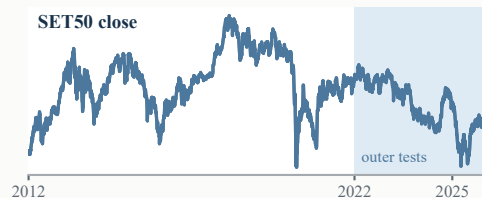


Point-in-Time Reliability Audit for Next-Day SET Index Direction Forecasting

SET50 2012-2025 • five frozen neural architectures • expanding out-of-sample evaluation

1 POINT-IN-TIME INPUTS



Dated financial news
2018-2025 • OOS sentiment

Five frozen architectures

windows • seeds • training budget

LSTM

CNN

LSTM-CNN

LSTM-Attention

LSTM-CNN-Attention

Target: next-session Up / Down

2 FIVE AUDIT DIMENSIONS

1

Temporal contract

Expanding windows • train-only transforms

2

Numerical denoising

Full-TA versus causal rolling VMD

3

Multimodal and LLM

Predicted news vs Market-Only

Bull LLM

Bear LLM

Leader

4

Regime and explainability

Bull/Sideway/Bear • train-only SHAP • LIME

5

Forward and transfer

Partial-2026 stress • frozen SET100

Matched cohorts • falsification controls • Holm adjustment • frozen transfer

3 AUDIT EVIDENCE

962 held-out sessions

2022-2025 • five fixed seeds

VMD: -0.60 to +0.35 pp

No conclusive directional effect

News: no robust BAcc gain

Leader: +5.93 to +6.00 pp

Intrinsic sentiment endpoint only

Regime-SHAP: CNN +1.46 pp

Mixed elsewhere • no Holm significance

LIME low fidelity: 71.83%

Forward failure modes detected

2026: one-sided collapse

SET100: weaker for 5/5 models

MAIN CONCLUSION

Apparent gains were architecture-dependent; no primary forecasting balanced-accuracy contrast survived the registered temporal, falsification, multiplicity and transfer checks.